Heat is not temperature

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Learning target: Distinguish temperature from energy transferred because of a temperature difference.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 20.*
No; it lacks 3 J of the required energy and remains bound under the model.

02. Worked example: explain the model
Same-temperature samples have no net heat flow due to a temperature difference; larger amount generally stores more internal energy for the same material/state.

03. Read and interpret the model
Temperatures equal; larger amount generally has more internal energy under comparable conditions.

04. Apply the physics
From block to water; block cools, water warms.

05. Explain your reasoning
Thermal energy transfers from hotter drink to cooler surroundings.

06. Review: connect an earlier idea *Review: Unit 4, Day 11.*
4 J of rotational kinetic energy. A rolling wheel also spins.

07. Exit check
Temperature in°C or K; heat transfer in J. Temperature is not an amount of heat.

08. STRETCH: test the model
No; need mass, material and state/reference for an energy comparison.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 20.*
New speed is 1/3; $v_{escape}=\sqrt{(2\ GM/r)}$.

02. Worked example: explain the model
Same-temperature samples have no net heat flow due to a temperature difference; larger amount generally stores more internal energy for the same material/state.

03. Read and interpret the model
Temperatures equal; larger amount generally has more internal energy under comparable conditions.

04. Apply the physics
Three times as much energy for the larger sample because $Q=mc\ \Delta T$.

05. Explain your reasoning
Thermal energy transfers from hotter drink to cooler surroundings.

06. Review: connect an earlier idea *Review: Unit 4, Day 11.*
$v=\sqrt{(2\ gh)}\approx 4$ m/s.

07. Exit check
Different amounts of material and different material/phase or microscopic energy structure.

08. STRETCH: test the model
40°C; equal heat capacities per sample make gains/losses symmetric. Unequal masses need an energy balance.